Advanced Programming Practice · 21CSC203P

Unit 3 · Session 9 · SLO 1 — Solved Worksheet

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Unit – 3 : Session – 9 : SLO – 1 | Describe MVC Architecture in GUI Development

I. Fill in the Blanks

1. View — the View is responsible for presenting the user interface.
2. Controller — it listens to user actions and modifies the Model.
3. Controller — an ActionListener attached to a button plays the Controller role in Swing MVC.
4. reusability — MVC promotes code reusability (separation of concerns) and testability.

II. True or False

1. FALSE — the Controller updates the Model; the View is refreshed from the Model, not updated directly by the Controller bypassing it.
2. TRUE — MVC separates data (Model), presentation (View) and input handling (Controller), which is the separation of concerns.

III. Multiple Choice

1. C — the Model contains and manages the application data and business logic.
2. C — the MVC control flow is View → Controller → Model → View.

IV. Programming

1. Identify the output: A — each button click increments the counter in the Model and the label is updated with the new value.
2. Identify the missing part: C — view.updateDisplay(model.getData()); after the Model changes, the View must be refreshed from the Model to complete the MVC cycle.